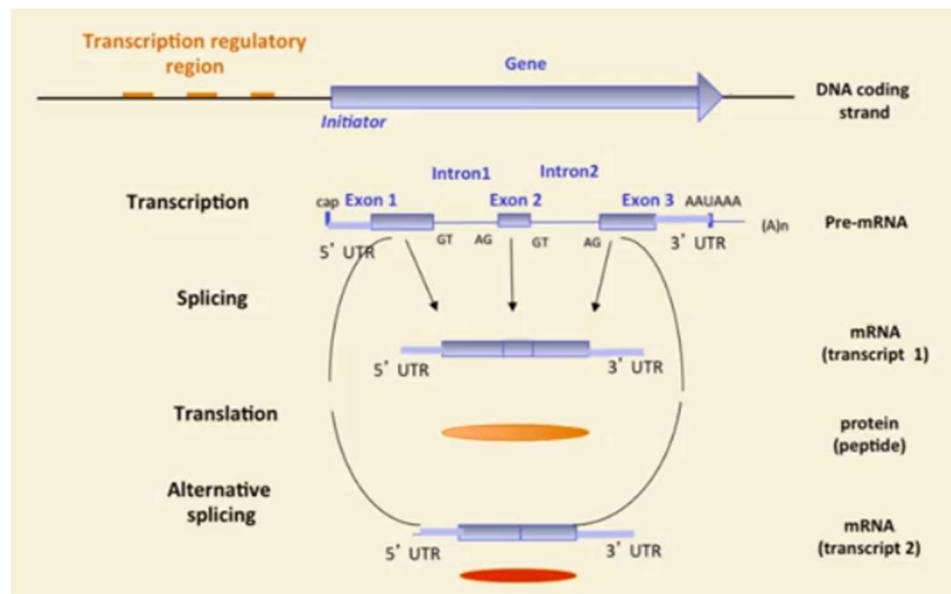


Genes and Transcripts

- Genes are composed of informative blocks (**exons**) interspersed with uninformative blocks (**introns**)
- Genes are housed in the genome and form a distinct body during gene expression
- Gene expression begins with transcription, where RNA polymerase creates a single-stranded pre-mRNA copy of the gene
- Pre-mRNA contains both exons and introns and is initially unstable in the nucleus.
- Pre-mRNA undergoes modifications for stabilization:
 - 5' capping
 - 3' cleavage and addition of a poly(A) tail
- Splicing, where introns are removed and exons joined together to form mRNA
- Mature mRNA exits the nucleus and travels to the cytoplasm for translation into protein
- One gene can produce one mRNA with multiple exons forming one protein
- Under different conditions (e.g., tissue type or health status), alternative splicing can occur, producing mRNAs with different exon combinations
- Exon skipping is when certain exons are excluded from the mRNA
- Genes can produce multiple splice variants, isoforms, or transcripts (terms used interchangeably)
- The example shows a simple case of alternative splicing with one mRNA missing exon 2

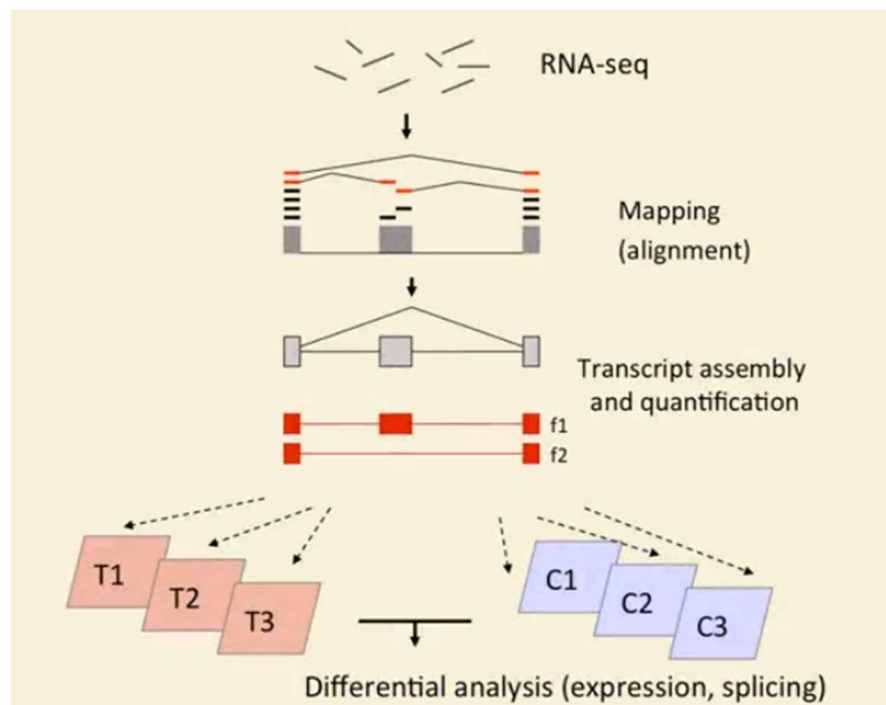


Questions Answered by Transcriptomics

- Assembly: What genes are expressed in the sample? What transcripts are expressed for each gene?
- Quantification: What are the genes' and transcripts' expression levels?
- Differential splicing and expression: How do expression levels and/or splicing patterns differ between two conditions (e.g. healthy vs disease tissue)?

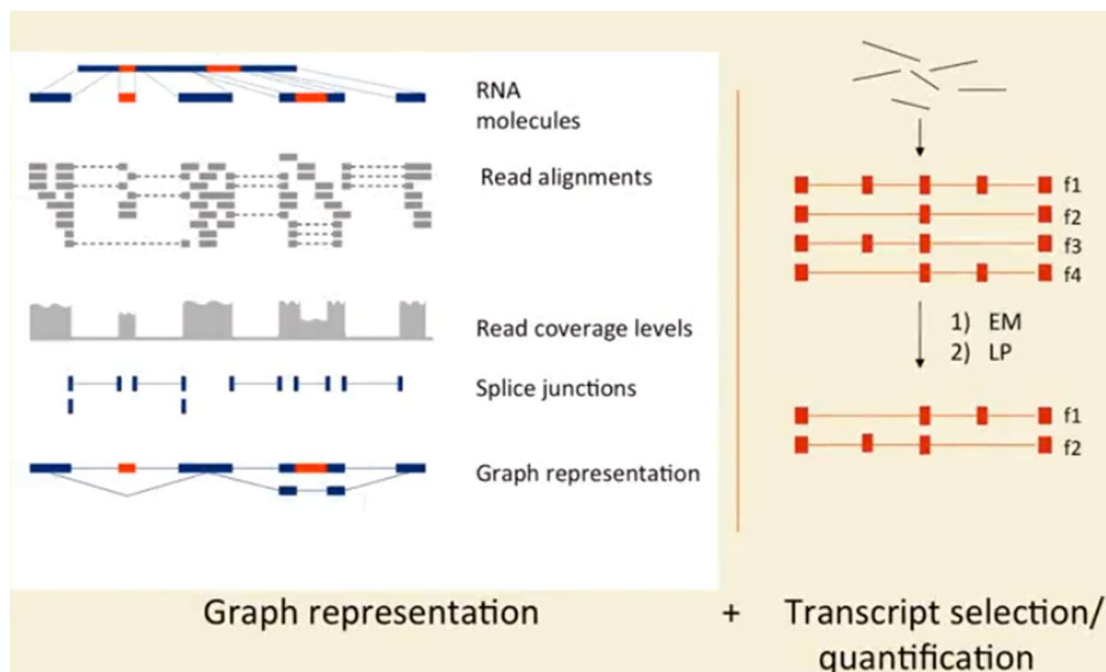
RNA-Seq

- RNA-seq experiments generate short reads (typically 50–250 base pairs long)
- These reads are mapped/aligned to the genome
- Aligned reads are clustered into gene-oriented groups to form basic gene and transcript structures, from which transcripts are decoded and quantified (expression levels are estimated)
- If there are two or more conditions (with or without replicates), analysis continues with differential expression analysis or differential splicing analysis between conditions



Transcript Assembly and Quantification

- Two key types of clues help identify gene structure:
 - **Bulk alignments** cluster in exonic regions; indicate exon locations
 - **Splice reads** span introns; indicate exon-exon junctions (introns)
- Assembly methods build a graph representation (exon graph) to capture possible transcript structures: Nodes = Exons and Edges = Introns (connections between exons)
 - Tracing **paths in the graph** (from start to end) yields possible **splice variants**.
 - Challenge: Graphs may encode **many more variants than are truly expressed**.



RNA-Seq Analysis Pipeline

1. Map reads to the genome (e.g., **Tophat***, **STAR**, **Hisat**)
- ↓
2. Assemble reads into transcripts (e.g., **Cufflinks***, **CLASS**, **iReckon**)
- ↓
3. Reconcile (partial) transcripts across multiple samples (**Cuffmerge***)
- ↓
4. Quantify isoform expression and compare among samples (e.g., **Cuffdiff***, **Ballgown**)